

Total No. of Questions : 8]

SEAT No. :

PC1926

[6353]-253

[Total No. of Pages : 3

T.E. (Robotics & Automation Engg.)

ROBOT PROGRAMMING

(2019 Pattern) (Semester-II) (311508 A)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Neat diagrams must be drawn wherever necessary.
- 2) Figures to the right side indicate full marks.
- 3) Use of calculator is allowed.
- 4) Assume Suitable data if necessary.

Q1) a) Explain the following instructions with relevant examples in VAL II: [10]

- i) PROMPT
- ii) APPRO
- iii) IOPUT
- iv) WSET
- v) ABORT

b) Write a short note on commands for listing, storing, retrieving program and location data in VAL II. [8]

OR

Q2) a) Write a pick and place VAL II language program for below given data: [9]

Assume speed percentage: 75

Pick coordinates: (355, 244, 390, -175, 9, -139)

Place coordinates: (355, 244, 390, -175, 9, -139)

End effector: Gripper

b) Write a short note production rate calculations using robot. [9]

P.T.O.

Q3) a) Explain the following instructions in detail: [10]

- i) SetDO do l5, 1;
- ii) MoveAbsJ p50, vl000, z50, tool2;
- iii) MoveL Offs(p2, 0, 0, 10), vl000, z50, tool1;
- iv) VAR num length;
- v) CONST num gravity := 9.81;

b) What are the differences between manual mode and automatic mode in ABB industrial robot? [8]

OR

Q4) a) Explain the programme structure of RAPID language. And explain the data types in RAPID language. [9]

b) Write a RAPID language code and flowchart to calculate the volume of cylinder. [9]

Q5) a) Write a short note on sensor commands in AML language. [9]

b) Explain the following instructions in AML language: [8]

- i) MOVE(<JX, JY, JZ>, <9.5, 1.4, 5.1>);
- ii) DEFIO(21, 0, 1, 1, 1);
- iii) READ(0, DATA1)
- iv) DISPLAY('Welcome', EOL);

OR

Q6) a) Write a short note on motion commands in AML language. [9]

b) Explain the following instructions with relevant examples in AML language: [8]

- i) ERASE
- ii) SETTLE
- ii) ACCEL
- iv) QGOAL

Q7) a) How is cycle time analysis conducted for robots, and what factors influence it? [9]

b) What are the key features of Robot Studio online software, and how are they utilized in different stages of robot programming? [8]

OR

Q8) a) How is collision detection implemented in robotics, and what methods are commonly used to prevent collisions? [9]

b) What challenges arise in scenarios involving multiple robots and machine interference, and how are they typically addressed? [8]

